التكنولوجيا التطبيقية
APPLIED TECHNOLOGY
وزارة التربية والتعليم والتعليم الفني

MINISTRY OF EDUCATION AND TECHNICAL EDUCATION
وزارة التربية والتعليم والتعليم الفنى

Physics question bank for the fourth formative test

Unit 12 : Modern electronics		الوحدات التدريبية
TPK03 – TPK04		أكواد المعارف
Unit 12: Modern electronics		المعارف
classification Of Elements According To Electric Conductivity And Doping With Donors And Acceptors.	TPK03	
The P-N Junction (Diode) And Bipolar Transistor (P-N-P) And (N-P-N) And Logic Gates.	TPK04	

Unit 12	Lesson one	Lesson two	Lesson three	Lesson four	Lesson five
Knowledge code	TPK03	TPK03	TPK04	TPK04	TPK04

المعرفة	الكود
Classification of elements according to electric conductivity and doping with donors and acceptors.	TPK03

1-True and false :

No	Question	True/False	W/R/NA
1	Semiconductors are materials that have an electrical conductivity between that of conductors and insulators.	T	
2	Silicon is the most commonly used semiconductor material	T	
3	Semiconductors can be doped with impurities to change their electrical properties	T	
4	The addition of a pentavalent impurity to a semiconductor creates a p-type material	W	
5	The addition of a trivalent impurity to a semiconductor creates an n-type material	W	

6	**Semiconductors are materials that have an electrical conductivity similar to that of conductors.**	W	

2-Choose the correct answer :

No	Question	A	B	C	D	Answer
1	Pure semiconductors behave as prefect ………at 0 K, whereas as temperature increases their conductivity increases.	Conductor.	Insulator.	Resistor	Capacitor	B
2	Silicon and germanium have a unique property in their electron structure each has …………… electrons in its outer orbit	5	3	2	4	D
3	……….. is one of the important and common elements in the universe.	Iron	Silicon	Aluminum	Indium	B

4	When the electron is freed, it leaves behind a vacancy in the broken bond (This vacancy is called a ………	Proton	neutron	hole	electron	**C**
5	The doping process It is the addition of a small percentage of some suitable …………. atoms in pure semiconductors.	Impurity	. proton	neutron	resistance	**A**
6	The doping process Increase the electrical ……………….... of semiconductors.	Resistivity	capacitance	magnetic	conductivity	**D**

3-Correct the underline:

No	Question	Answer
1	Semiconductors are materials that have an electrical conductivity **similar** to that of conductors	**different**
2	Silicon is the **only** commonly used semiconductor material	**not the only**
3	Doping a semiconductor with impurities **has no effect on** its electrical properties	**changes**
4	The addition of a **pentavalent** impurity to a semiconductor creates a p-type material.	**trivalent**
5	The addition of a **trivalent** impurity to a semiconductor creates an n-type material.	**pentavalent**
6	**Decrease** the temperature is one of the methods to increases the electric conductivity of semiconductors.	**increase**

4- Mention the scientific term:

No	Question	Scientific term
1	It is the addition of a small percentage of some suitable impurity atoms in pure semiconductors	Doping of Semiconductor
2	It is an extrinsic semiconductor which is obtained by doping with trivalent impurity atoms	p-type semiconductor
3	It is an extrinsic semiconductor which is obtained by doping with pentavalent impurity atoms	n-type semiconductor
4	The number of bonds broken per second will be equal to the number of bonds mended per second **OR** Fixed number of free electrons and free holes remains constant at every temperature	Dynamic equilibrium (thermal equilibrium)
5	It is a regular arrangement of atoms in the solid state	The crystal
6	They are materials with conductivities somewhere between conductors and insulators	Semiconductors

5- Give a reason for each of the following:

1- At very low temperature, pure silicon crystal is an insulator

Because all bonds (covalent bonds) in the crystal are unbroken (there are no free electrons)

2- By adding some impurity atoms, the conductivity of semiconductors will increase

When a pure semiconductor is doped with impurity atoms (donor or acceptors), the charge carries (electrons or holes) will increase, therefore the conductivity of semiconductors will increase due to movement the charge carries.

6- Mention the necessary conditions to obtain:

No	Question	Answer
1	P – type	The pure semi conductor is doped with trivalent atoms such as Boron

2	N – type	The pure semi conductor is doped with pentavalent atoms such as phosphors
3	Increasing the electrical conductivity of pure semiconductor	1) Increasing the temperature 2) Pure semiconductor is doped with impurity atoms (pentavalent or trivalent)

المعرفة	الكود
The P-N Junction (Diode) And Bipolar Transistor (P-N-P) And (N-P-N) And Logic Gates	**TPK04**

1-Choose the correct answer:

No	Question	A	B	C	D	Answer
1	**What is a p-n junction?**	**A junction between two p-type semiconductors**	**A junction between a p-type and an n-type semiconductor**	**A junction between two n-type semiconductors**	**A junction between a metal and a semiconductor**	**B**
2	**Which semiconductor material has a higher concentration of holes, p-type or n-type?**	**p-type**	**n-type**	**Both have the same concentration of holes**	**None of the above**	**A**
3	**What is the depletion region in a p-n junction?**	**The region with a high concentration of free charge carriers**	**The region with a low concentration of free charge carriers**	**The region with a uniform concentration of free charge carriers**	**The region with a non-uniform concentration of free charge carriers**	**B**

4	What happens to the depletion region under forward bias?	It increases in width	It decreases in width	Its width remains constant	It disappears	B
5	What is the purpose of a p-n junction?	To allow current to flow in only one direction	To amplify signals	To reduce noise in circuits	To increase the speed of computer processors	A
6	Which of the following is a characteristic of a p-n junction?	High resistance in both directions	Low resistance in both directions	High resistance in forward bias and low resistance in the reverse bias	Low resistance in forward bias and high resistance in the reverse bias	D
7	What is the basic construction of an NPN transistor?	Three layers of p-type semiconductor	Two layers of p-type semiconductor and one layer of n-type semiconductor	Two layers of n-type semiconductor and one layer of p-type semiconductor	Three layers of n-type semiconductor	C
8	Which region of an NPN transistor is the emitter?	The middle region	The region closest to the negative terminal of the battery	The region closest to the positive terminal of the battery	It depends on the bias voltage applied to the transistor	B
9	What is the function of the base in an NPN transistor?	To provide a path for the flow of current	To amplify the signal	To control the flow of current between the emitter and the collector	To increase the voltage of the signal	C
10	Which of the following is the correct equation for the current gain of an NPN transistor?	$\beta = I_C / I_B$	$\beta = I_B / I_C$	$\beta = I_E / I_C$	$\beta = I_C / I_E$	A
11	Which logic gate has an output that is the inverse of its input?	AND gate	OR gate	NOT gate	XOR gate	C
12	What is the output of an AND gate if both inputs are 1?	0	1	Undefined	It depends on the specific configuration	B

					of the AND gate	
13	What is the output of an OR gate if both inputs are 0?	0	1	Undefined	It depends on the specific configuration of the OR gate	A
14	What is the output of a NOT gate if the input is 0?	0	1	Undefined	It depends on the specific configuration of the NOT gate	B

2-True and false :

No	Question	True/False	W/R/NA
1	In a p-n junction, the p-type material has an excess of electrons, while the n-type material has a minority of electrons.	F	
2	The depletion region in a p-n junction is an area that is depleted of free charge carriers.	T	
3	When a p-n junction is forward-biased, the depletion region becomes wider	F	
4	When a p-n junction is reverse-biased, the depletion region becomes narrower	F	
5	P-n junctions are used in a wide range of electronic devices, including diodes, transistors, solar cells, and light-emitting diodes (LEDs).	T	
6	In an NPN transistor, the emitter is doped with a higher concentration of impurities than the collector	T	
7	- In a NPN transistor, the base is doped with a higher concentration of impurities than the emitter	F	
8	In an NPN transistor, the majority of the current flows from the emitter to the collector.	T	

9	Logic gates are used to manipulate and process digital signals, but not analog signals.	T	
10	The NOT gate has a single input and a single output.	T	
11	The output of an AND gate is true only when all of its inputs are true.	T	
12	The OR gate produces a true output when any of its inputs are true.	T	
13	- The NAND gate is a combination of an AND gate followed by a NOT gate.	T	
14	- The NOR gate is a combination of an OR gate followed by a NOT gate.	T	
15	A logic gate can have more than two inputs.	T	
16	AND, OR, and NOT are the only types of logic gates.	F	
17	A NOR gate produces a low output when any of its inputs are high.	T	
18	A NAND gate produces a high output when any of its inputs are low.	T	

3-Solve the following :

No.	Question	Answer

1	Draw a transistor circuit as a switch at on condition then calculate the collector current of the transistor. A lamp having a resistance of 22 Ω is to be switched by a BJT. The power supply voltage (Vcc) is 5V and the collector emitter voltage is 0.1V. Calculate the minimum base current required to place the transistor in on condition if beta is 80	$V_{cc} = I_c R_c + V_{ce}$ $5 = I_c (22) + 0.1$ $I_c = 0.22273A$ $I_c = \beta I_b$ $I_b = 0.00278A$
2	In an NPN transistor 108 electrons enter the emitter in 10–8 s. If 1% electrons are lost in the base, calculate the current gain (the electron charge (e) = 1.6×10^{-19} C)	Ne =It N: no. of electrons $10^8 \times 1.6\times10^{-19} = I_e (10^{-8})$ $I_e = 1.6\times10^{-3}A$ $I_b = 0.01\times1.6\times10^{-3}A$ $I_c = 0.99\times1.6\times10^{-3}A$ $I_c = \beta I_b$ $0.99\times1.6\times10^{-3} = \beta\times0.01\times1.6\times10^{-3}$ $\beta = 99$
3	A transistor is connected in Common Emitter configuration. The voltage drop across the load resistance (RC) 3 k Ω is 6 V. Find the base current. α of the transistor is 0.97	$V_c = I_c R_c$ $6 = I_c \times 3000$ $I_c = 0.002A$ $\beta = \frac{\alpha_e}{1-\alpha_e} = \frac{0.97}{1-0.97} = 32.333$ $\beta = \frac{I_C}{I_B}$ $I_B = \frac{0.002}{32.333} = 6.185 \times 10^{-5}A$

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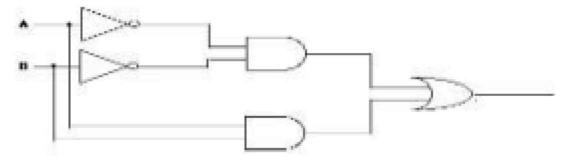
<td>4</td>
<td>The electronic circuit below represents combining gates for a special job. Complete the following truth table for the following circuit

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<td><table><tr><th>A</th><th>B</th><th>Output</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table></td>
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<td>5</td>
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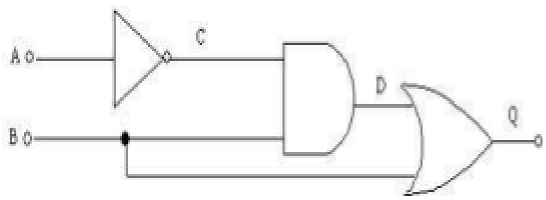
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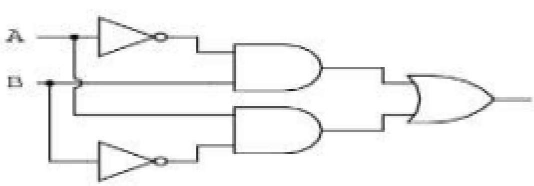
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</tr>
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4- Give a reason for:

1- Emitter in a transistor is heavily doped

Because it emits charge carriers through the thin base layer to be collected by the collector

2- The thickness of the base of the bipolar junction transistor must be very small

Because very small quantity of electrons which emits from emitter is lost to recombination inside base, whereas, the rest of electrons get conducted through collector.

3- Diode is used to a switch

Because the current flows in only one direction which is closed in the forward connection (conducting) and open (non- conducting) in the reverse connection

4- Digital electronics is preferred to analog electronics

Because the digital electronics have high ability to overcome the electrical noise and its basic idea is to code information in binary form (0 or 1)

5- Mention one function or application for each of the following:

No	Question	Answer
1	the transistor	1)Amplifier 2)switch
2	logic gates	1)Computer circuits 2)calculators, digital watches, computers, robots, industrial control systems, and in telecommunications

6- Mention the scientific term:

No	Question	Scientific term
1	The p-type of the diode is connected to the positive terminal and the n-type is connected to the negative terminal of a battery	Forward Connection (forward bias)
2	The N-type is connected to the positive terminal and the P-type is connected to the negative terminal of the battery	Reverse Connection (Reverse bias)
3	It is the ratio between the collector current and the base current	Current gain (βe)
4	The electronics which deals with natural quantities	Analog electronics

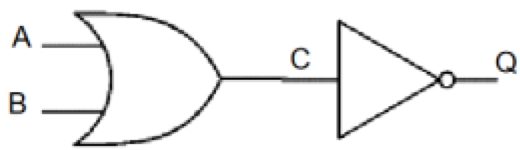
5	A new branch of electronics has developed, such that the signal is in terms of one of two possible values representing two states 0 or 1but is coded.	**Digital electronics**
6	These are the circuits that perform logic operations	**Logic Gates**

7- Correct the underline word:

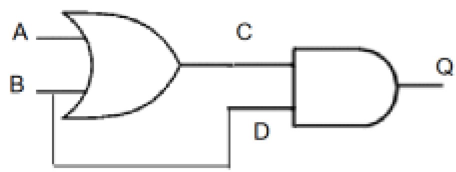
No	Question	Answer
1	The depletion region in a p-n junction is an area that is **rich in** free charge carriers.	**low in**
2	The **forward** biasing of a p-n junction increases depletion region and prevents current from flowing through the junction.	**reverse**
3	The **reverse** biasing of a p-n junction decreases the depletion region and allows current to flow through the junction	**forward**
4	Doping a semiconductor involves adding impurities to **hold** its electrical properties.	**change**
5	N-type semiconductors are doped with impurities that have fewer valence electrons than the semiconductor crystal	**more**
6	P-type semiconductors are doped with impurities that have more valence electrons than the host material.	**less**
7	Doping a semiconductor with a **trivalent** impurity creates an n-type material.	**pentavalent**
8	Doping a semiconductor with a **pentavalent** impurity creates a p-type material.	**trivalent**
9	The doping process is the **only** way to alter the electrical properties of a semiconductor material.	**one of the ways**
10	A **PNP** transistor consists of a p-type semiconductor sandwiched between two n-type semiconductors.	**NPN**
11	In a NPN transistor, the **base** is doped with a higher concentration of impurities than the collector.	**emitter**
12	In a NPN transistor, the **emitter** is doped with a lower concentration of impurities than the collector.	**collector**
13	The DC current gain of a NPN transistor is defined as the ratio of the **emitter** current to the base current.	**collector**

14	A logic gate can have one or more inputs and <u>**one or more**</u> outputs	one

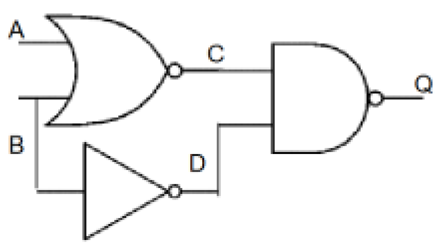
<u>8- Solve the following:</u>



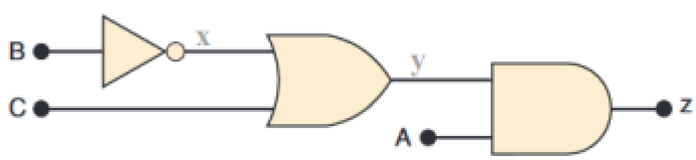
A	B	C	Q
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0



A	B	C	D	Q
0	0	0	0	0
0	1	1	1	1
1	0	1	0	0
1	1	1	1	1



A	B	C	D	Q
0	0	1	1	0
0	1	0	0	1
1	0	0	1	1
1	1	0	0	1



A	B	x	C	y	Z
0	0	1	0	1	0
0	0	1	1	1	0
0	1	0	0	0	0
0	1	0	1	1	0
1	0	1	0	1	1
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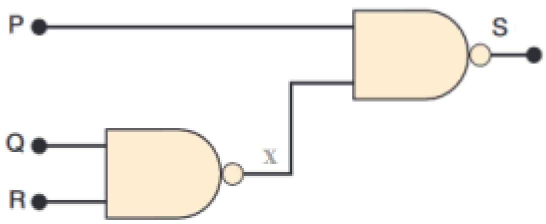
1	1	0	0	0	0
1	1	0	1	1	1



P	Q	R	x	S
0	0	0	1	**1**
0	0	1	1	1
0	1	0	1	**1**
0	1	1	0	**1**
1	0	0	1	0
1	0	1	1	**0**
1	1	0	1	**0**
1	1	1	0	**1**